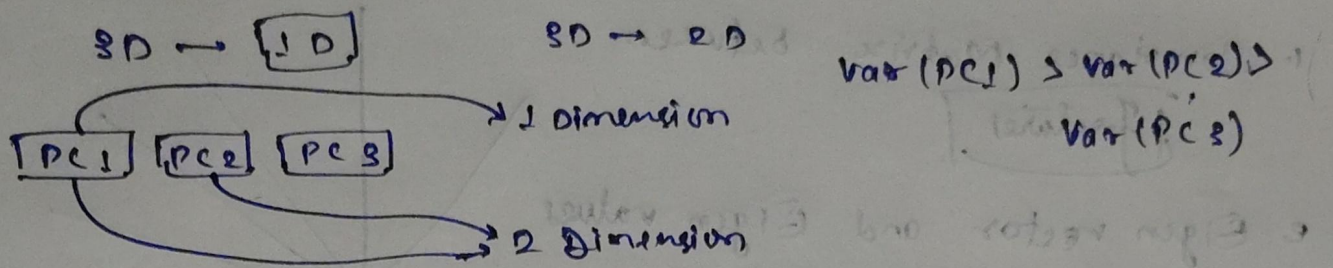
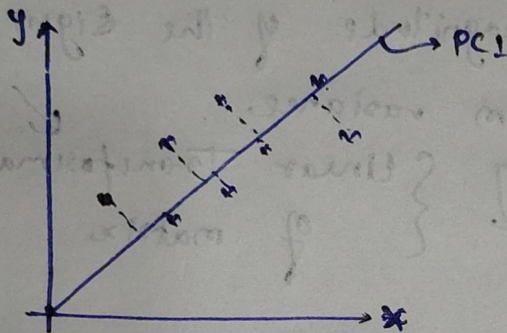


To get the best principle component which captures maximum variance

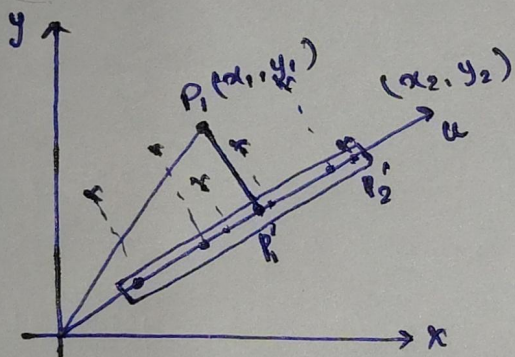


* Math Intuition behind PCA Algorithm.



2D $\rightarrow$ 1D

- ① Projection
- ② cost function $\rightarrow$ variance



$$\text{Proj}_{P_1} u = \frac{P_1 \cdot u}{\|u\|}$$

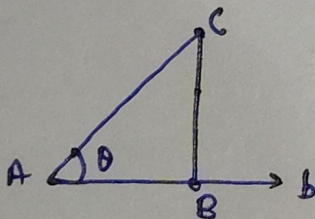
$\|u\| \rightarrow$ 1 unit vector

$$\text{Proj}_{P_1} u = P_1 \cdot u \Rightarrow \text{Scalar value}$$

$\downarrow$
 P_1'

$$[P_0', P_1', P_2', P_3', P_4' \dots P_n']$$

Scalar value
 $\downarrow$
variance



$$[P_0', P_1', P_2', P_3', P_4' \dots P_n']$$

$\downarrow$

$$x_0', x_1', x_2', x_3', x_4' \dots x_n'$$

Max Variance = $\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$

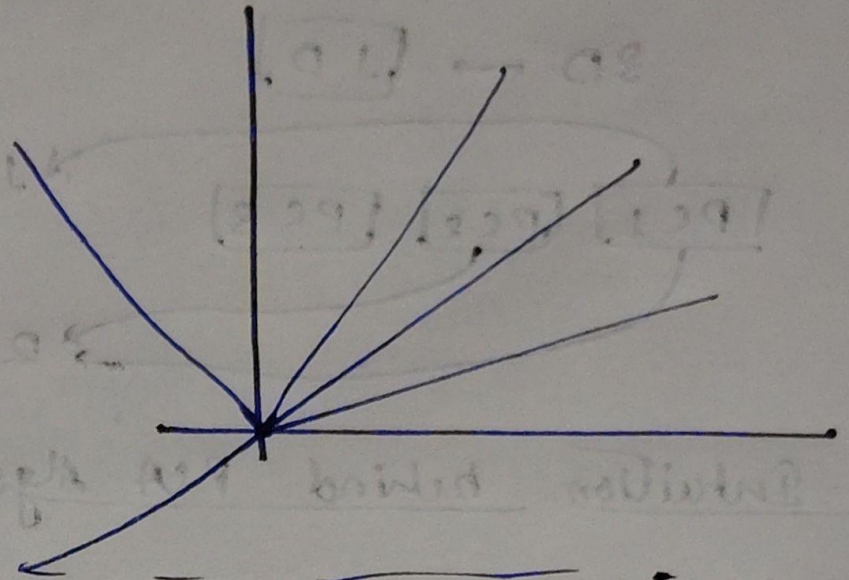
$\downarrow$
cost function

Goal: Find the best unit vector which captures maximum variance

Eigen value Vector and Eigen value

1. Covariance Matrix between
Features

2. Eigen vector and Eigen values
will be found out from this
covariance matrix.



3. Eigen vector $\rightarrow$ Eigen value $\rightarrow$ magnitude of the Eigen

vector $\rightarrow$ capture the maximum variance.

$$AV = \lambda V$$

{ Linear Transformation
of matrix }